Capital Bikeshare Implementation Report

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I. Data Management and Implementation

A. Data Gathering

The major part of the system data has been captured from the official website of Capital Bikeshare located in Washington DC.

The bike and the overall fleet information currently being used is generated using the AI tool, Mockaroo.

To work with the market expansion of the bikeshare service, cycle counter information has been collected from Bike Arlington which is one of the counties participated in this Capital Bikeshare venture.

The data is collected for the period between 2021 and 2023 to understand user behaviour on consuming the service, environmental impact, to analyse the bike usage statistics to identify the requirement of funding to perform regular maintenance. Bicycle and pedestrian counter data is collected and mocked to simulate a year data to identify hotspots for future station proposals.

B. Data Implementation

The data is gathered, cleaned and is ready to be inserted to SQL server database.

The database design involves segregating the data collected to be inserted to 6 relational tables having a column with primary key associated with other tables in the database.

Visualisation of this dataset is done using Tableau for creating visual reporting and dashboards to identify insights.

II. System and Database Implementation

A. Tools Used

- *SQL Server*: The choice of a relational database selected is Microsoft SQL Server due to the large dataset collected for the analysis.
- *Tableau*: This visualisation tool is picked due to its ease of use and diverse visualisation capabilities and flexibility with data exploration. It has a large and active community which was beneficial during this project.

B. Database Design

For the analysis, 6 relational tables are created, and the necessary primary and foreign key constraints are implemented to the database. Necessary constraints are added to check and prevent any junk or miscellaneous character being inserted into the table.

Following are the data definition of the tables created in SQL server database.

LAPTOP-OOR2TCCN\...e - dbo.Bike_Info

Column Name	Data Type	Allow Nulls
Bike_ID	numeric(18, 0)	☐
Bike_Type	nvarchar(50)	☑
Purchase_date	datetime	☑
Purchase_Price	float	☑
Distance_Travelled	float	☑
Maintenance_status	nvarchar(50)	☑
last_maintenance_date	datetime	☑
last_maintenance_cost	float	☑
past_maintenance_cost	float	☑
last_station_ID	numeric(18, 0)	☑

LAPTOP-OOR2TCCN\S...are - dbo.Billing

Column Name	Data Type	Allow Nulls
Bill_ID	numeric(18, 0)	☐
Bill_Date	datetime	☑
Bill_Status	nvarchar(50)	☑
Payment_Type	nvarchar(50)	☑
[Distance travelled]	float	☑
Cost	float	☑

LAPTOP-OOR2TCCN\...e - dbo.Customer

Column Name	Data Type	Allow Nulls
Customer_ID	numeric(18, 0)	☐
Age	numeric(18, 0)	☑
Gender	nvarchar(50)	☑
member_Casual	nvarchar(50)	☑
Subscription_plan	nvarchar(100)	☑
Plan_Expiry	datetime	☑

LAPTOP-OOR2TCCN\....Ride_Transaction

Column Name	Data Type	Allow Nulls
Ride_ID	nvarchar(50)	☐
Rideable_Type	nvarchar(50)	☑
[Started At]	datetime	☑
[Ended At]	datetime	☑
[Start Station ID]	numeric(18, 0)	☑
[Start Station Name]	nvarchar(150)	☑
[End Station ID]	numeric(18, 0)	☑
[End Station Name]	nvarchar(150)	☑
Customer_ID	numeric(18, 0)	☑
Start_lat	float	☑
Start_lng	float	☑
End_lat	float	☑
End_lng	float	☑
Bill_ID	numeric(18, 0)	☐
Bike_ID	numeric(18, 0)	☑

LAPTOP-OOR2TCCN\...are - dbo.Station

Column Name	Data Type	Allow Nulls
Station_ID	numeric(18, 0)	☐
Name	nvarchar(150)	☑
Latitude	float	☑
Longitude	float	☑
Num_Bikes_Available	numeric(18, 0)	☑
Num_Docks_Available	numeric(18, 0)	☑
Num_Ebikes_Available	numeric(18, 0)	☑
Capacity	numeric(18, 0)	☑

LAPTOP-OOR2TCCN\...dbo.Station_Info

Column Name	Data Type	Allow Nulls
Object_ID	numeric(18, 0)	☐
Station_ID	numeric(18, 0)	☑
[Station Status]	nvarchar(150)	☑
[Has Kiosk]	numeric(18, 0)	☑
[Rental Methods]	nvarchar(150)	☑

Fig. 1. Tables created in MS SQL Database

III. Dashboards

Dashboards are created in Tableau to visualize and identify insight based on the analytical requirements discussed in the specification document. The dashboards built provides insights on user behaviour analysis, station utilization metrics, financial performance analysis, competitor analysis, maintenance predictive analysis and for market expansion.

A. *User Behaviour Analysis*

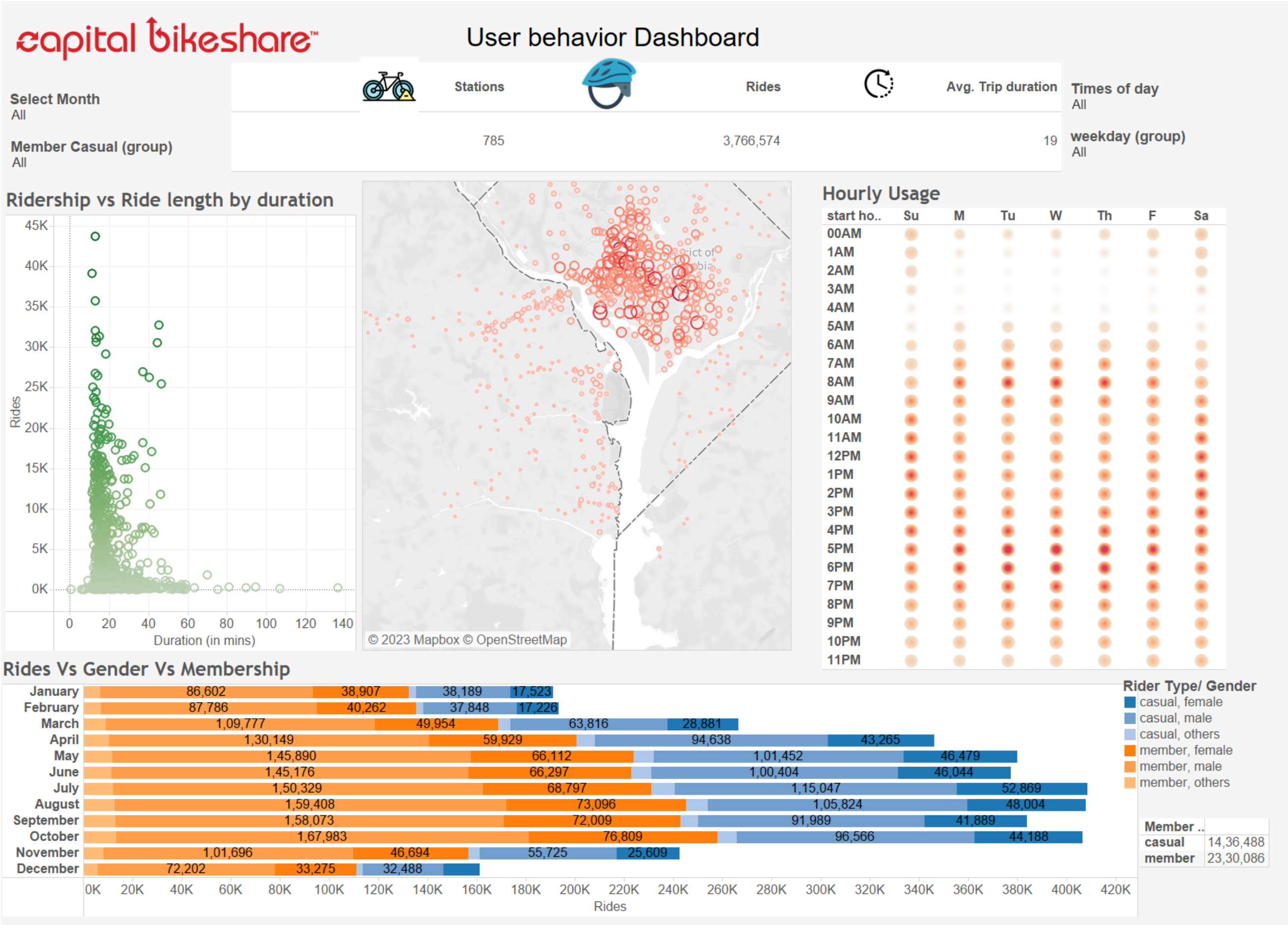

Fig. 2. User Behaviour Dashboard

Two dashboards are created to understand the nature of customer who uses the bikeshare service and following insights are derived to customers in Washington area based on the data collected.

This is a comprehensive dashboard to understand the customer behaviour that are using the Bikeshare service in Washington D.C. The dashboard shows few analyses by which the customers use the service to commute to different location in the capital. The data captured during login and during the ride help to understand the peak usage pattern over the days and by hours (Hourly Usage), the hotspot locations where the number of rides were higher compared to other stations in the area based on month and weekends. Rides Vs Gender Vs Membership chart to identify the distribution of rides over the months and classify by whether the customer is of type member or causal and a split by Gender. It is evident that the peak time of service usage is between 6am and 10am and between 3pm and 7pm. Similarly, the usage pattern changes on weekends where the peak usage is between 10am and 7pm. The dashboard has additional filters to calculate the average duration of the trips based on a selection of stating stations from the map palate available in the dashboard. Membership riders are contributing to maximum rides over the months in 2022-2023. There is a seasonal slowness identified between November and March where the number of rides is lower compared to other months. This dashboard summarises that there are 785 active stations that are operating and has a total ride of close to 38 million and an average ride time of 19 mins overall. The dashboard has the option to filter

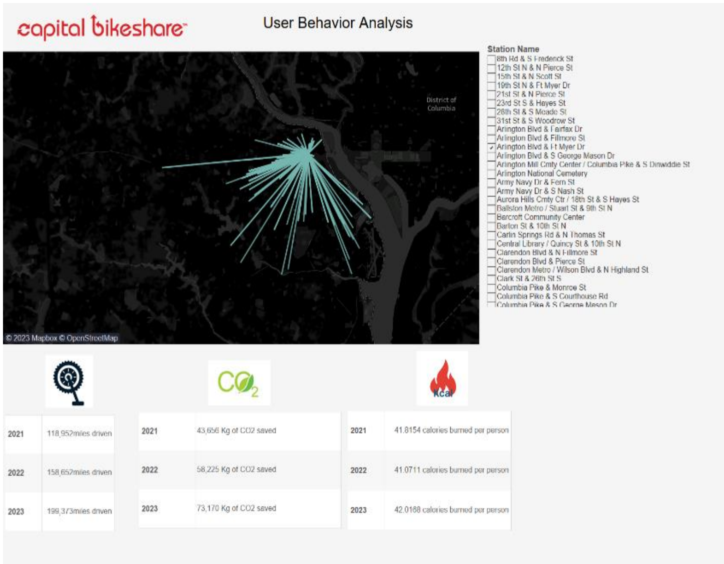

Fig. 3. User Behaviour Dashboard

the data and users can use this dashboard to identify insights related to customer behaviour for any time of the day, based on gender, rider type and from popular stations around the city.

The dashboard in fig 3. helps to understand the user behaviour and trackers on few health and environmental parameters that are captured. Customers use bikeshare as an alternative for low-cost commutability and also, they are environmentally conscious to take bikeshare as the option to reduce the emission of toxic gases due to combustion. Few users use bike to go on a trail around the city and might travel a farther distance compared to the usual. This dashboard helps to identify the possible destinations from a particular start station and provide insights on the average time taken to reach the specific destination. This dashboard is implemented by using the geo spatial coordinates between the start location and the end location and find the distance travelled based on a small calculation.

The dashboard helps to identify the connecting stations and helps to gain insights on stations which are predominantly used and can help to decide possible future station establishment locations.

B. *Bike Utilization Metrics*

This comprehensive dashboard helps to understand the Capital bikeshare fleet on the number of availability of bikes in each category that are currently in streets. Based on the insights captured, decision can be made for scheduling maintenance of bike for services and allocate further bikes in case of any shortage. It also has a tracker on the current count of availability of bikes that are running without any issue, count of bikes that are scheduled for maintenance and also for bikes that are currently in service and might not be available for customer usage.

Capital bikeshare has a total of approximately 6000 bikes in the roads and are providing service in three segments such as Classic bikes, docked bikes and E-bikes. Each bike in the fleet has a maintenance cost associated with for the bikes that are purchased between 2016 and 2022. The density chart helps to identify the amount of cost incurred on the previous scheduled maintenance for the different type of bikes.

The dashboard has a bar chart, which captures the insights on the total distance travelled by each bike during their usage and help the business to identify the bikes that require a scheduled maintenance so that the company can provide fit bikes to the customers for their usage.

This dashboard also provides the information about how

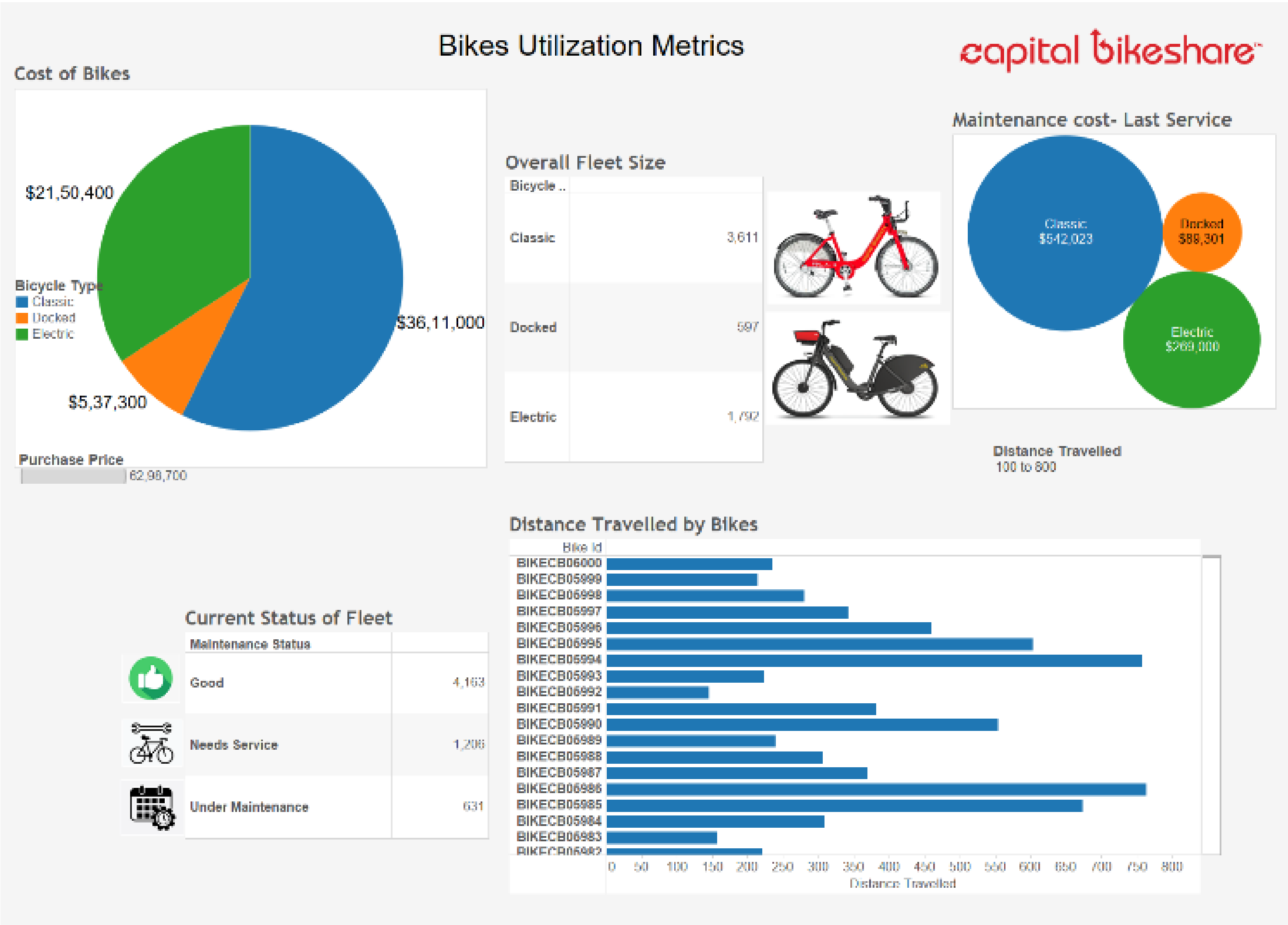

Fig. 4. Bikes Utilization Dashboard

much amount invested on each segment of bikes and could verify that Capital bikeshare has invested heavily in Classic bikes as they cover the larger part of the fleet and the next being the E-Bikes, which is technologically advanced and target the audience for a quicker commute in between the cities.

C. Station Utilisation Metrics

This dashboard will help to identify the top 10 most popular and the least popular stations in the regions selected. where the demand for the bikes is high. This dashboard helps to plan the portability of bikes from one station to other based on the number of docks and the bikes available at each station. Business can get an insight on the time of peak usage, to increase more riders, they can deploy and circulate bikes from the sparingly used stations to much busier stations.

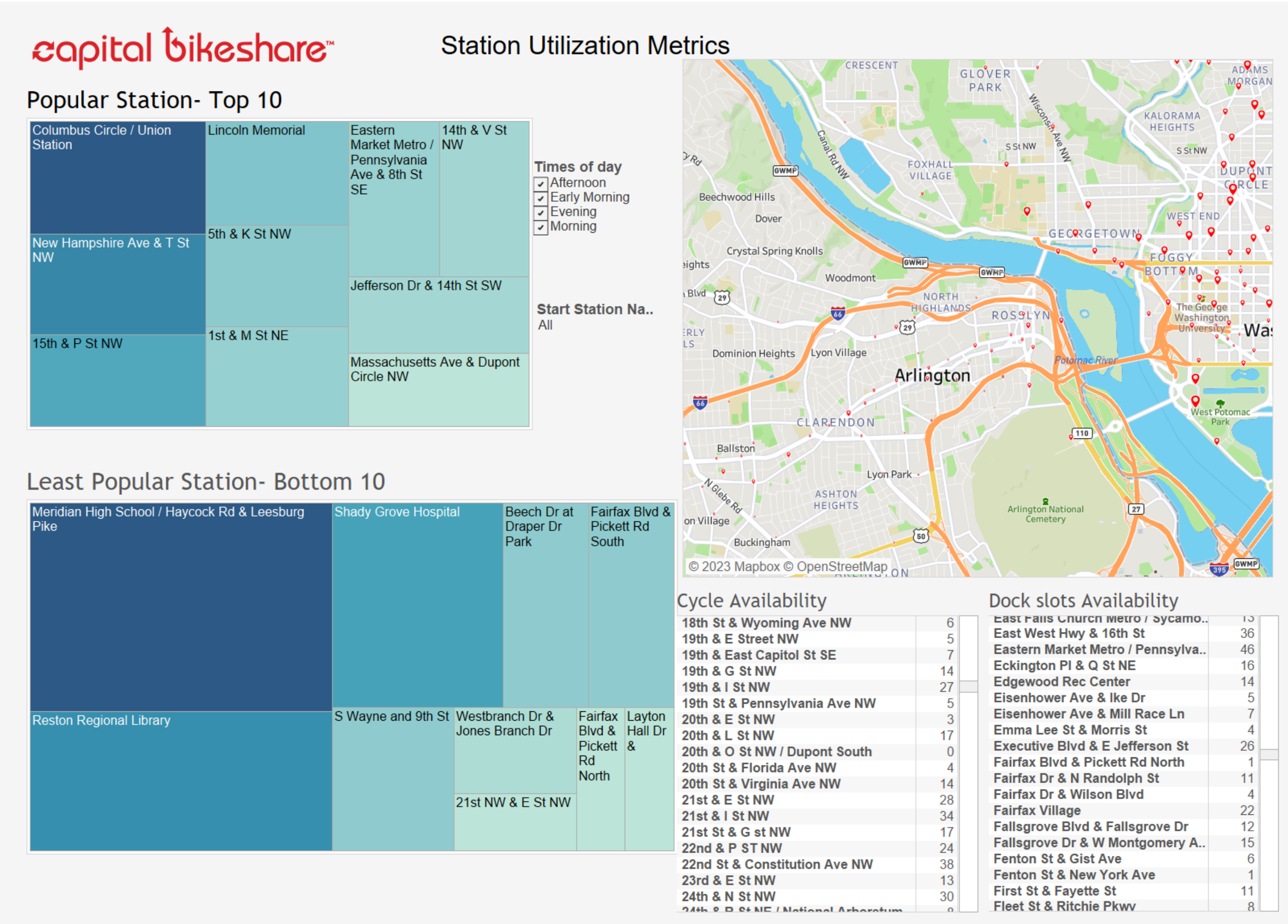

Fig. 5. Station Utilisation Dashboard

From the overall regions operated by the Capital bikeshare, few of the top 10 popular stations with maximum rides comes from the middle of the city – Columbus Circle/ Union Station. This is followed by New Hampshire Ave and by Lincoln memorial. Looking at the data, major hotspot of the popular stations are located in the city centre and they are closer to historical and important places in the city. Fair share of the rides will be due to sight-seeing and other attractions near to these places and tourists who visits the city are choosing the bikeshare service to commute to these sight - seeing.

Majority of the casual riders might be tourists in these areas as they are visiting and staying in Washington for a shorter period, and it is evident that the major user pattern comes from these monumental places.

Least popular station is calculated using the heatmap based on the fewer trips to these stations and business can decide to circulate the bikes from these locations to much busier places.

D. Financial Performance Analysis

Major part of the revenue comes from the membership program that is in place in Capital Bikeshare and further by the duration and the distance travelled in the bike by members and casual riders.

In between October 2022 and October 2023, there is an estimated revenue of close to 29 million. In the revenue analysis section, the dashboard helps to understand the source of the revenue generated. Analysis is based on user membership or due to casual riding of the bikes, based on Age, whether which group of customers uses the bikes most and over the months in 2023.

Based on the data captured on the bike used in each trip and the distance travelled, classic bike holds the major share as it brings Capital bikeshare more than 75% of the revenue. This is followed by the latest electric bikes which accounts for 5 million in revenue. Docked type bikes are sparingly used over the years and generated a very small audience.

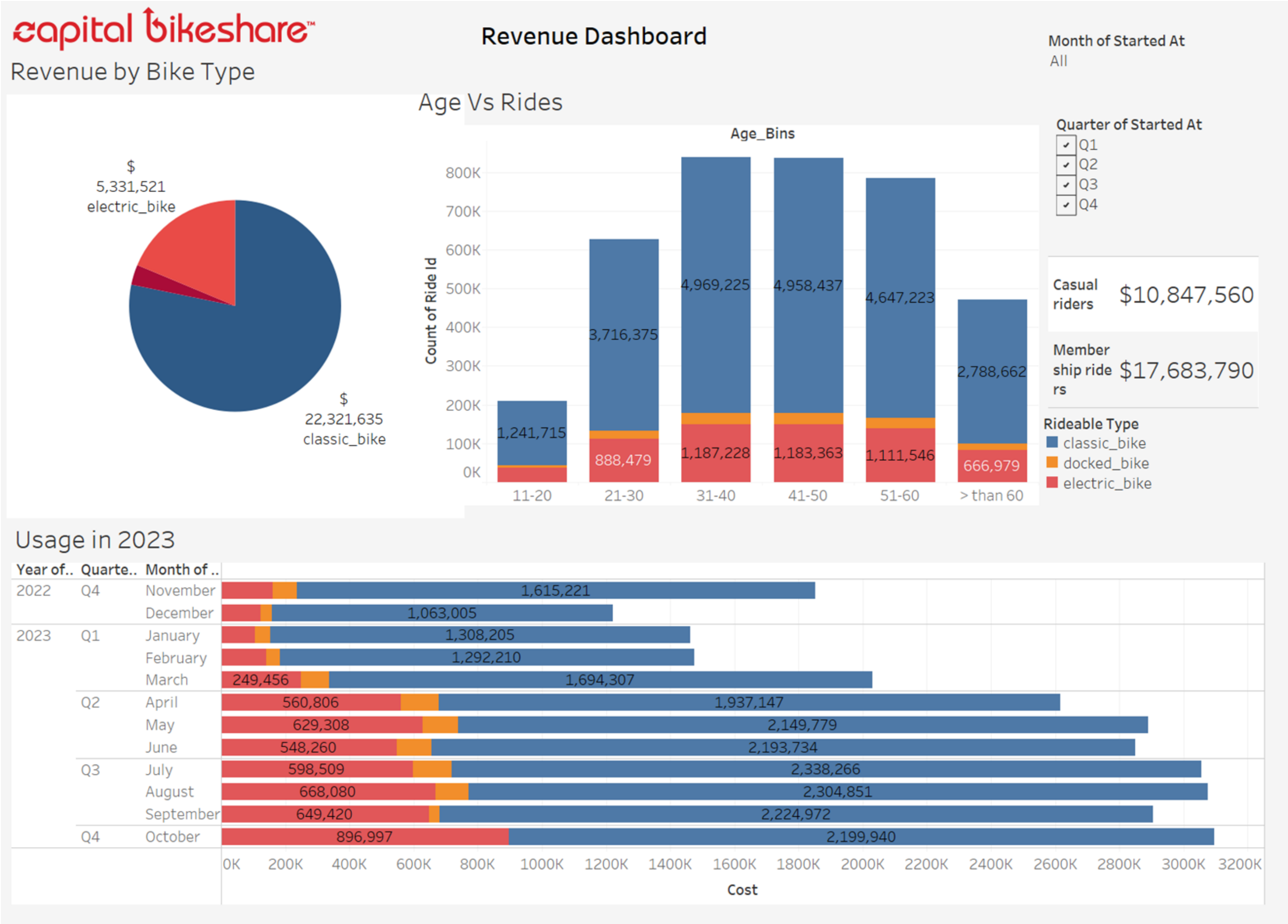

Fig. 6. Financial Performanace Analysis

Analysis based on the quarter wise revenue over the last year shows an increasing trend which also shows that the classic bike is popular among the Capital bikeshare customers. The business is showing a down trend during the holiday season and the business picks backs up starting from February. There is also an uptrend in the electric bikeshare usages over the months and has contributed to 20% of the revenue.

On further analysis based on the age wise over the revenue shows people aged in between 30 and 55 accounts for the majority in the revenue and there is a comparatively lower revenue generated by younger generation. Further investigation needs to be done for better conclusion on the lower usage by young customers.

There is a stable revenue generated close to an average 5 million for users using classic bikes for age between 30 and 60. This comprehensive dashboard will help to narrow down all the features mentioned earlier and understand better insights on the revenue generated.

E. Competitor Analysis

The membership data captured, helps to understand the distribution of the bike service usage in Washington area and Capital bikeshare is facing a stiff competition between Lime and Divvy bikes where the Lime bikeshare scheme has a better membership pattern over the years and Divvy, another bike sharing company is approaching at a consistent rate. Better business decisions need to be taken to bring capital bikeshare.

Better membership plans targeting the right audience and effective marketing of the plans available to the customers will bring the capital bikeshare to a better position. Innovation and technology advancement also needs to be considered to significantly be a competitor in the market.

To understand the internal process, survey information on the bike usage is used to find any discrepancies in the firm. The data is collected from 1000 respondents and were asked to rate the service provided by the capital bikeshare and the dashboard shows an average rating of close to 3 in bike rating, customer service rating and satisfaction rating on a scale of 5.

Marketing of the bikeshare company needs to be done to increase the average rate of all the parameters. Referral programs, presence in social media and conducting partner program with local government can increase the popularity of Capital bikeshare among the customers.

Fig. 7. Competitior Analysis

F. Market Expansion Analysis

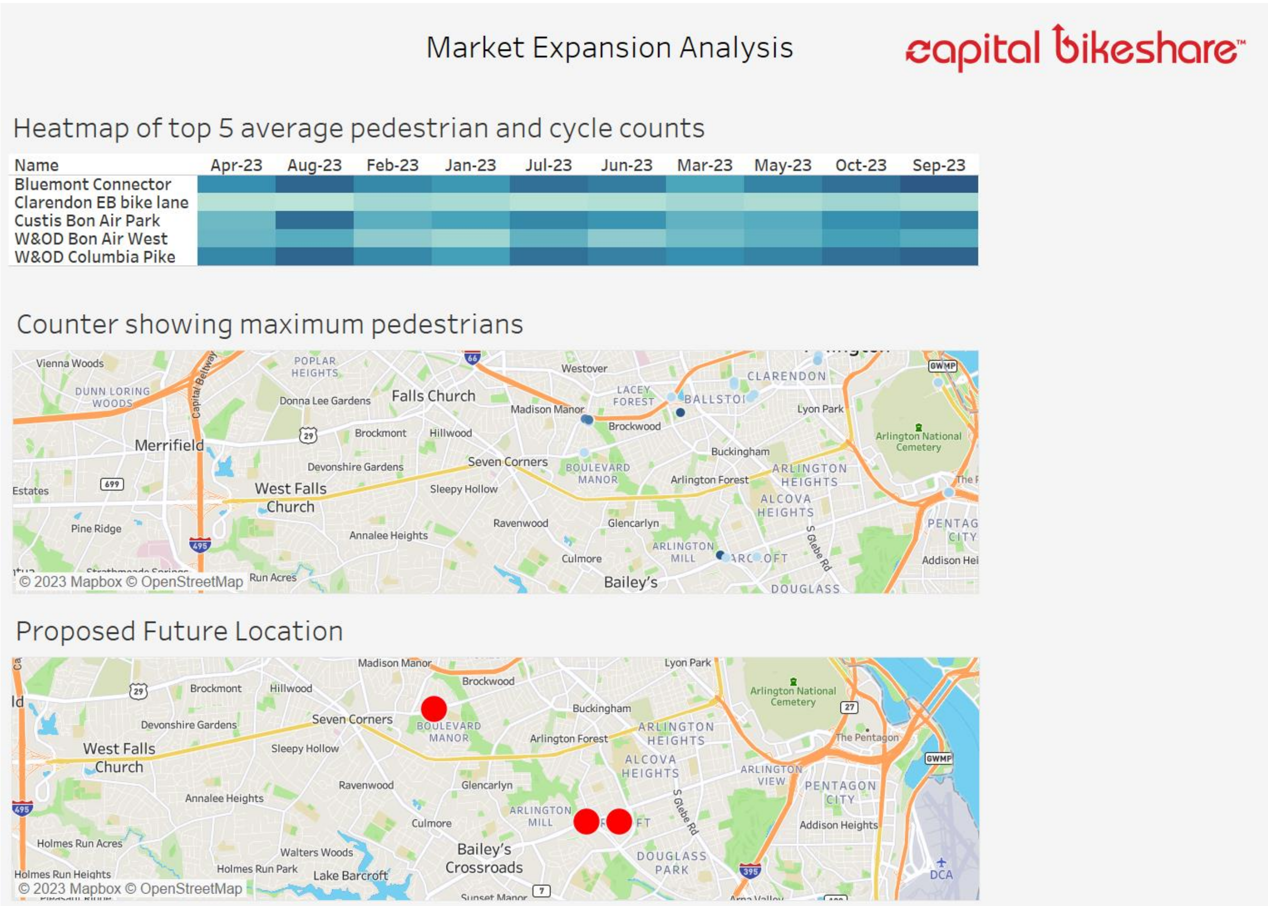

Fig. 8. Market Expansion Dashboard

Counter information of pedestrians and cycle usage in specific area is captured and used to analyse and able to find the hotspot to establish a new station as part of the market expansion. Top 5 location having higher footfall presence were identified and suitable new station locations were identified based on survey conducted to customers.

Based on the counter information collected, three of the highest footfall spots are identified.

Bluemont Connector, Custin Bon Airpark, W&OD Colombia Pike. They have an average footfall of close to 10000 per hour and be an ideal place for setting up of station.

Stations having least ridership needs to be relocated to other busier spots for better presence and increase in the use of the bikeshare service.

Supply of bikes to busier location needs to planned using station analysis for effective balance of bikes in the station having busier customer presence.

Survey needs to be conducted to identify the location for future bike stations to understand and address the need for this transportation service.

IV. Conclusion

The most attractive expansion option both in the service changes and other station expansion. Since for shorter distance commute enables customers to use capital bikes for the last mile connectivity, presence of stations closer to one another so that in busier station, customer can still reach the nearby station to unlock a bike to travel thereby reducing the waiting time.

Few recommendations given by the analysis is the increase of docks or stations in the existing stations.

More stations near the residential neighbourhood as bikeshare will be an ideal option for the last-minute commute to the destination. Expansion to area where capital bikeshare don't operate now.

More presence in commercial and employment area will attract newer customers. Due to the continuous environmental awareness of customers, they tend to use the bikeshare service due to its lower environmental impacts.

Providing adequate consideration in increasing the electric bikes as to adapt new technological advancement in this transportation method so that commuting is made easy to customers at their convenience.

Since the pricing is based on the time and distance travelled, users having certain membership plans can provide with longer free use period.

To check on the feasibility of using the charging station for the bikeshare electric bikes to charge private bikes for a smaller price to make a revenue when the electric bikes are in the streets. Analysis needs to be done to find the ideal time for charging purpose in the current stations by running a pilot scheme in specific routes to study the user behaviour.

Marketing needs to be done effectively to increase the customer membership. Price needs to be adjusted in a competitive way so that new members join the plan and be part of the family. Increase in the funding of promotion and marketing campaigns to increase the ridership and enrolment in the membership plans.

To encourage young riders to use capital bikeshare, membership plan or pass needs to be tailored by giving free rides by a referral program for students and discounted membership plans to use capital bikeshare.

Have corporate plans and stations near industrial centres for customers to use bikeshare for their commute.

Increase the electric and classic bikes around tourist attractions as they contribute to the fair share of the casual riders in the city.

Introduce and promote the impact of riding bikes as an alternative for health and environment benefits.

V. Challenges and Limitations

A. *Infrastructure Requirements*

Having a comprehensive distribution of bike lanes around the city and parking facilities is highly essential for the integration of bike sharing systems. Proper planning and consideration need to be done to cater the bikeshare service scheme to the larger customer for their use.

B. *Funding and Sustainability*

For a continued operation, securing a long-term regular funding and developing sustainable business models are challenges for bike sharing programs. Currently the Capital Bikeshare rely on the local government and federal grants for their operational expenses. More public-private partnership needs to be added to sustain in the competitive market in the future for providing better service.

C. *Safety Concerns*

Cyclists and motorists need to be properly educated about the rules to be followed and to increase the awareness of the safety of the riders to mitigate the risk of accidents, ensuring the wellbeing of the riders and promoting the bikeshare scheme.

D. *Limited Accessibility*

Majority of the concentration of the stations is located in the city centre and the availability of the bikeshare scheme is lower in residential neighbourhoods. This will hinder the accessibility of the bikeshare as a mode of transportation to the users far from city. Proper planning and distribution of stations around the city

E. *Technology Advancement*

Latest technology needs to be integrated with this bike sharing service so that the industry is relevant to the rapidly evolving advancements in the technology. Research and development need to be carried out and sufficient funding needs to be allocated to find newer technology integration to the bike sharing sector.

References

[1] System data from https://capitalbikeshare.com/ (last accessed on 06/12/2023)

[2] Cycle counter information on https://www.bikearlington.com/ (last accessed on 06/12/2023)

[3] Membership data from https://data.bts.gov/stories/s/Bikeshare-and-e-scooters-in-the-U-S-/fwcs-jprj/ (last accessed on 06/12/2023)